

Circuit Analysis (Lab # 2)

Experiment 2: Measurements of Waveforms and Fundamental Quantities, Basic resistive circuits.

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Grade and Signature:

CLO3	Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)			Total marks (10)	

Objectives: This experiment will provide exposure to measurements of waveforms and Fundamental Quantities, Basic resistive circuits. A primary purpose of this lab course is for you to master the use of electronic test equipment and measurement of fundamental quantities.

Equipment required:

Power Supply, Breadboard, Oscilloscope, Signal Generator Connecting wires, Resistances, Multimeter. **Procedure**

Experiment # 1

SIGNAL GENERATOR AND OSCILLOSCOPE

1. Use the signal generator and the oscilloscope to perform the following tasks.
2. Build circuit 'A' shown below in Figure 1.
3. Set the signal generator to generate a sinusoidal signal with a frequency of 100 Hz and peak-to peak voltage of 5V.
4. Set up one probe across input signal, and another across R1 and R2 alternatively.
5. Measure and draw all the voltages and time related parameters on the graphs below.

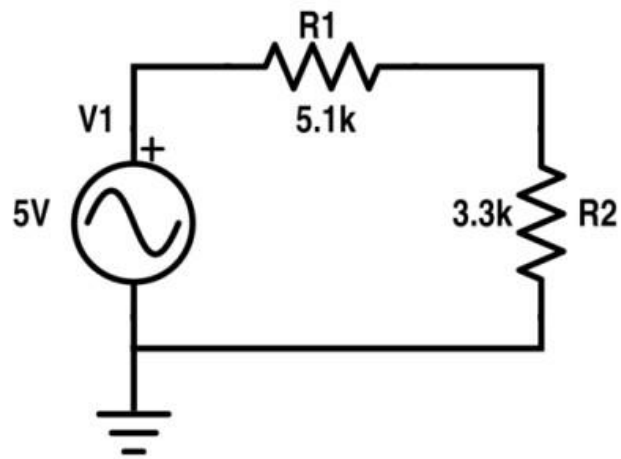


Figure 1 Circuit 'A'

Resistor	Input voltage	RMS Voltage	P-P Voltage
R1	5 V	1.07 V	3.036 V
R2	5 V	0.694 V	1.964 V

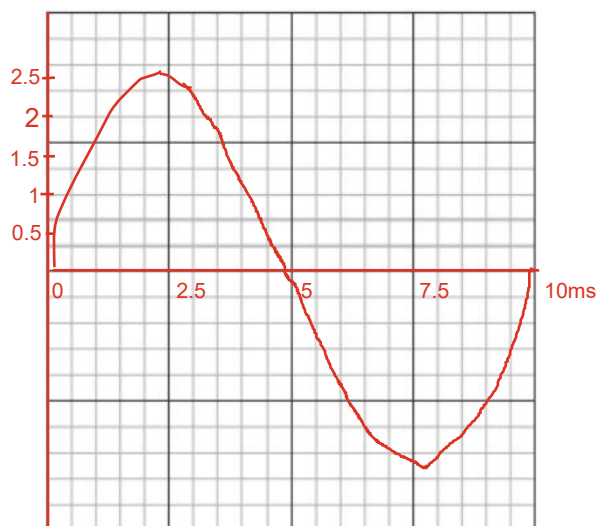


Figure 2 Input Voltage Signal

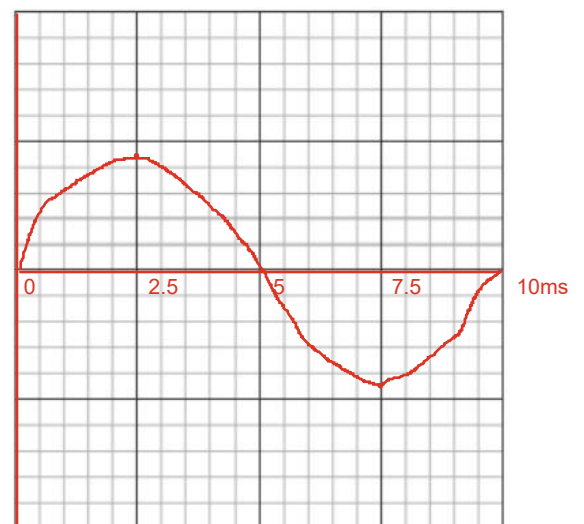


Figure 3 Voltage Signal across R1

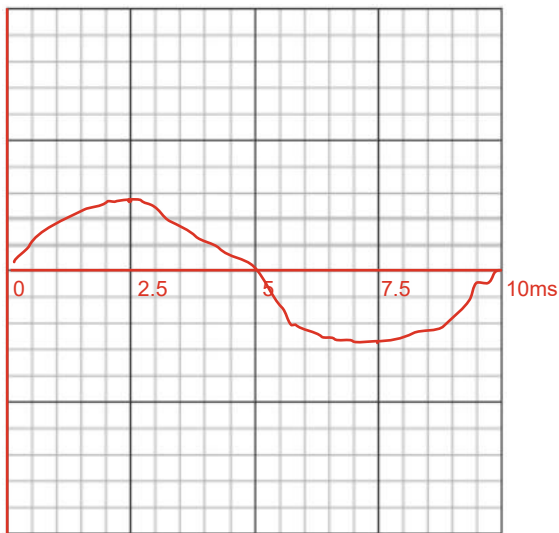


Figure 5 Voltage Signal across R2

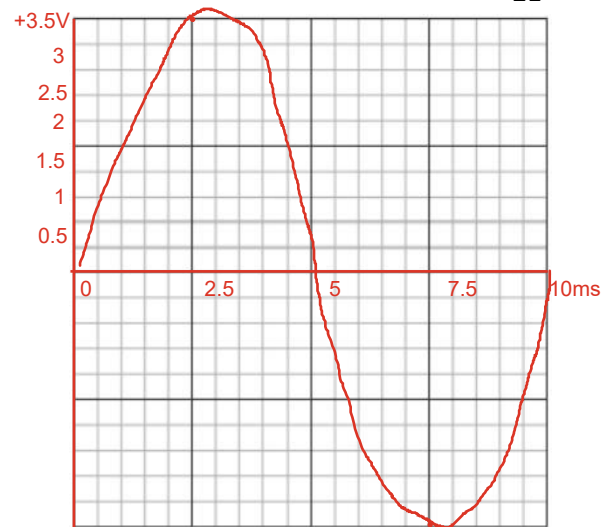


Figure 6 Draw 7V Sinusoidal

Experiment # 2

SIGNAL GENERATOR AND OSCILLOSCOPE

1. Use the signal generator and the oscilloscope to perform the following tasks.
2. Build circuit 'B' shown below in Figure 7.
3. Set the signal generator to generate a sinusoidal signal with a frequency of 200 Hz and peak-to-peak voltage of 7V.
4. Set up one probe across input signal, and another across R1, R2 and R3 alternatively.
5. Measure and draw all the voltages and time related parameters on the graphs below.

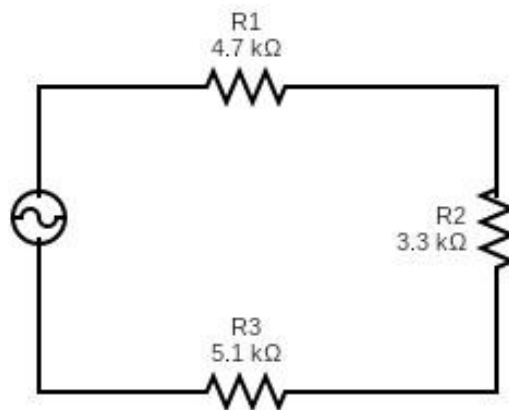


Figure 7 Circuit 'B'

Resistor	Input voltage	RMS Voltage	P-P Voltage
R1	7 V	0.89 V	2.51 V
R2	7 V	0.62 V	1.76 V
R3	7 V	0.96 V	2.73 V
R1 and R2	7 V	1.51 V	4.27 V
R2 and R3	7 V	1.59 V	4.49 V

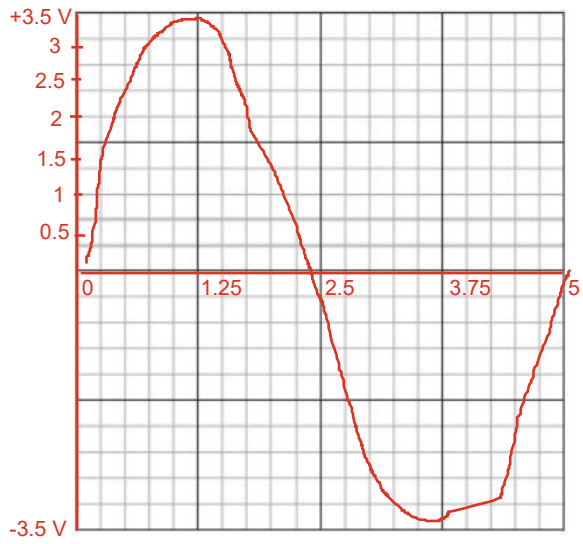


Figure 8 Input Voltage Signal

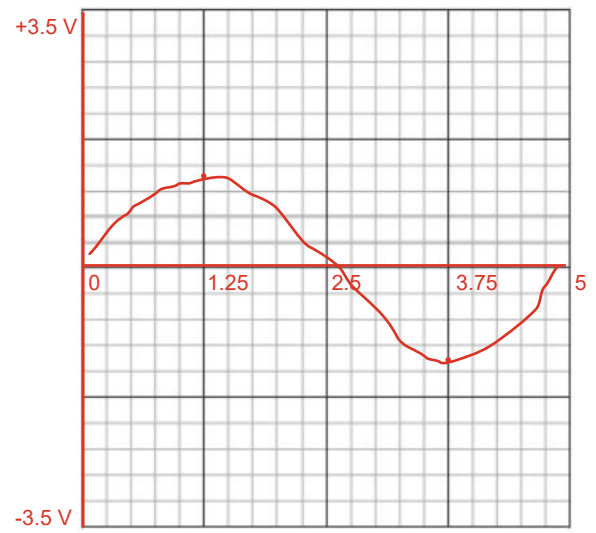


Figure 9 Voltage Signal across R1

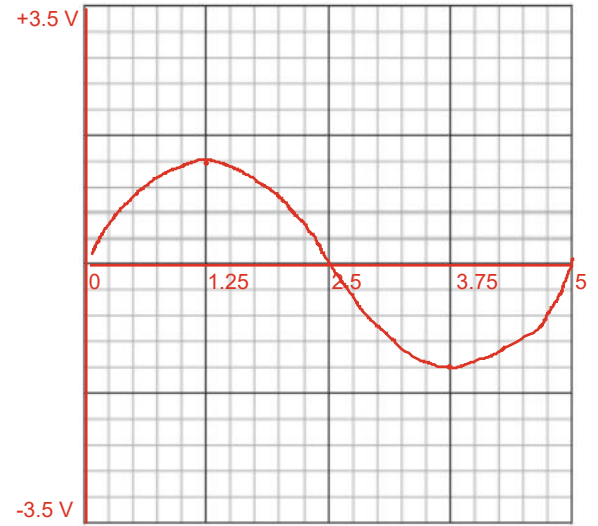
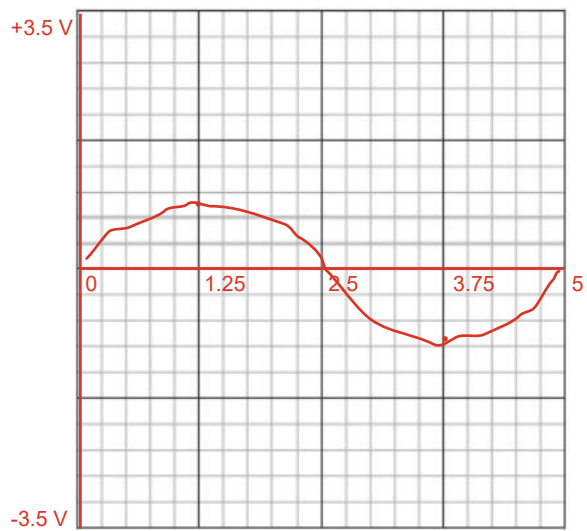


Figure 10 Voltage Signal across R2

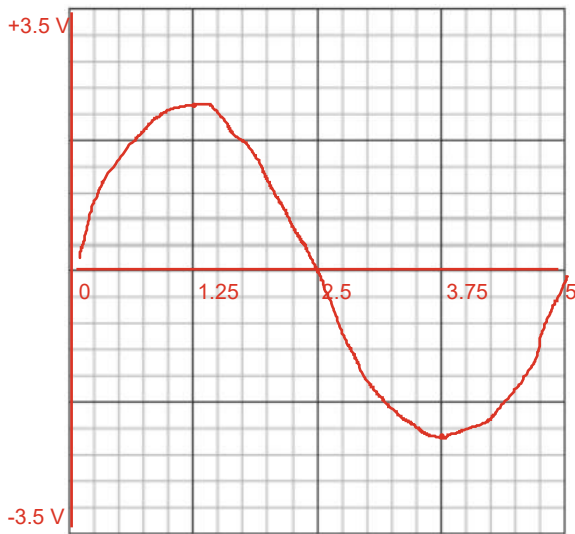


Figure 11 Voltage Signal across R3

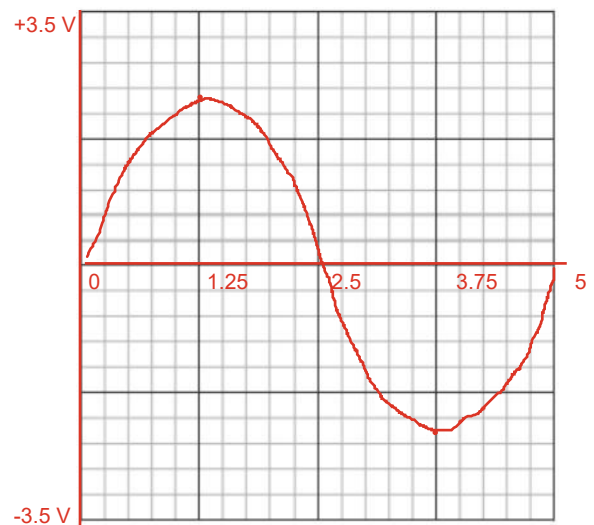


Figure 12 Voltage Signal across R1 and R2

Figure 13 Voltage Signal across R2 and R3

Post Lab Questions

1. Define Ohm's Law.

Ohm's Law states that the voltage (V) across a conductor is directly proportional to the current (I) flowing through it, provided the temperature and physical conditions remain constant. Mathematically, $V = I \times R$, where R is the resistance measured in Ohms (Ω). This means if resistance is constant, doubling the voltage doubles the current. Ohm's Law is the foundational principle of electric circuit analysis.

2. What is a resistor?

A resistor is a passive two-terminal electrical component designed to oppose the flow of electric current and dissipate electrical energy as heat. It is characterized by its resistance value, measured in Ohms (Ω). Resistors are manufactured using resistive materials such as carbon, metal film, or wire-wound elements. They are widely used in circuits to limit current, divide voltages, set gain levels, and protect sensitive components.

3. How does a resistor work?

A resistor works by impeding the movement of electrons through its resistive material. When current flows through a resistor, free electrons collide with the atoms of the material, losing kinetic energy which is released as heat (Joule heating: $P = I^2R$). The resistance value depends on the material's resistivity (ρ), its length (L), and cross-sectional area (A), given by $R = \rho L/A$. A higher resistance allows less current for the same applied voltage, in accordance with Ohm's Law.

4. What is your inference from the input and output waveforms on oscilloscope?

From the oscilloscope waveforms, the following inferences are drawn:

(a) The output waveforms across all resistors remain sinusoidal at the same frequency as the input. Resistors are linear elements and do not introduce phase shift or frequency distortion.

(b) The amplitudes across individual resistors are smaller than the input, confirming the voltage divider action. For Circuit A: $V_{R1} \approx 3.04 \text{ Vpp}$ and $V_{R2} \approx 1.96 \text{ Vpp}$, which add up to 5 V (KVL verified).

(c) In Circuit B, R2 and R3 are in parallel, so they share the same voltage across their terminals. The voltage across the parallel combination is smaller than across R1, consistent with the calculated ratio of their equivalent resistance to total resistance.

5. Show calculations of 7V input signal (Figure 6) from signal generator.

a. Voltage Calculations (following volt/div and number of boxes on oscilloscope):

Setting: Volt/div = 1 V/div

Observed peak of waveform = 3.5 divisions above centre line

Peak Voltage (V_{peak}) = divisions $\times$ volt/div = $3.5 \times 1 = 3.5 \text{ V}$

Peak-to-Peak Voltage (V_{pp}) = $2 \times V_{\text{peak}} = 2 \times 3.5 = 7 \text{ V}$ ✓

RMS Voltage = $V_{\text{peak}} / \sqrt{2} = 3.5 / 1.4142 \approx 2.475 \text{ V}$

b. Frequency / Period Calculations (following steps in frequency measurement):

Signal Generator set frequency: $f = 200 \text{ Hz}$

Period: $T = 1/f = 1/200 = 0.005 \text{ s} = 5 \text{ ms}$

Setting: Time/div = 1 ms/div

One complete cycle spans: 5 divisions on oscilloscope

Verification: $T = 5 \text{ div} \times 1 \text{ ms/div} = 5 \text{ ms} \rightarrow f = 1/T = 1/5 \times 10^{-3} = 200 \text{ Hz}$ ✓

c. Current Calculation using Ohm's Law (Circuit B, 7V source):

Given: $R_1 = 4.7 \text{ k}\Omega$, $R_2 = 3.3 \text{ k}\Omega$, $R_3 = 5.1 \text{ k}\Omega$

Step 1 — Equivalent resistance of parallel combination ($R_2 \parallel R_3$):

$$R_{\text{eq}} = (R_2 \times R_3) / (R_2 + R_3) = (3300 \times 5100) / (3300 + 5100) = 16,830,000 / 8400 \approx 2002.5 \Omega$$

Step 2 — Total circuit resistance:

$$R_{\text{total}} = R_1 + R_{\text{eq}} = 4700 + 2002.5 = 6702.5 \Omega$$

Step 3 — Peak source voltage: $V_{\text{peak}} = V_{\text{pp}} / 2 = 7 / 2 = 3.5 \text{ V}$

Step 4 — Peak total current (by Ohm's Law):

$$I_{\text{peak}} = V_{\text{peak}} / R_{\text{total}} = 3.5 / 6702.5 \approx 0.5222 \text{ mA}$$

Step 5 — RMS current:

$$I_{\text{rms}} = I_{\text{peak}} / \sqrt{2} = 0.5222 / 1.4142 \approx 0.369 \text{ mA}$$

Step 6 — Current through individual branches:

$$I_{R1} = I_{\text{rms}} = 0.369 \text{ mA} \text{ (R1 is in the main series branch)}$$

$$\text{Voltage across parallel combination: } V_{\text{eq}} = I_{\text{rms}} \times R_{\text{eq}} = 0.369 \times 2002.5 \approx 0.739 \text{ V (rms)}$$

$$I_{R2} = V_{\text{eq}} / R_2 = 0.739 / 3300 \approx 0.224 \text{ mA (rms)}$$

$$I_{R3} = V_{\text{eq}} / R_3 = 0.739 / 5100 \approx 0.145 \text{ mA (rms)}$$

$$\text{Check: } I_{R2} + I_{R3} = 0.224 + 0.145 = 0.369 \text{ mA} = I_{R1} \text{ ✓ (KCL verified)}$$